

OS PRACTICAL

PRACTICAL-2

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

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#rootect(0x731f4ed81000, 4896, PROT_READ) = 0
#rootect(0x731f4ed81000, 4896, PROT_READ) = 0
#rootect(0x0004af100000, 162722, PROT_READ) = 0
#rootect(0x731f4e320000, 8192, PROT_READ) = 0
printout(0, RLIMIT_STACK, NULL, (rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY)) = 0
getuid(0) = 0
getgid(0) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x8baf14bf, 0xfab6d851]},
f_namelen=0, f_frsize=4096, f_flags=0, f_val=[0x1f180b1d, 0xf180b1d], f_fsid={val=[0x1f180b1d, 0xf180b1d]}}) = 0
statfs("/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x8baf14bf, 0xfab6d851]},
f_namelen=0, f_frsize=4096, f_flags=0, f_val=[0x1f180b1d, 0xf180b1d], f_fsid={val=[0x1f180b1d, 0xf180b1d]}}) = 0
brk(NULL) = 0x6040bbf43000
brk(0x6040bbf43000) = 0x6040bbf43000
open(AT_FDCWD, "/proc/filesystems", 0_RDONLY|O_CLOEXEC) = 3
fcntl(3, {it_mode=3_FMODE, it_size=0}, 0) = 0
read(3, "node[types]vnode/vtaps/vnode/vtd...", 1024) = 442
seek(3, 0, SEEK_CUR) = 442
seek(3, 416, SEEK_SET) = 0
close(3) = 0
open(AT_FDCWD, "/proc/mounts", 0_RDONLY|O_CLOEXEC) = 3
fcntl(3, {it_mode=3_FMODE, it_size=0}, 0) = 0
read(3, "none /usr/lib/modules/6.18.33-2-... 1024) = 1024
read(3, "e 0 /devpts /devpts defaults rw,... 1024) = 1024
read(3, "accesslist,mixed=SS36,trans,... 1024) = 1024
read(3, "one /run/credentials/systemd-res... 1024) = 769
read(3, "... 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(0, F_GETFD) = 0
fcntl(3, F_GETFD) = 0
fcntl(3, F_GETFD) = 0
sigaction(SIGTSTP, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
poll([fd=0], events=0, [fd=1, events=0], [fd=2, events=0], 3, 0) = 0 (timeout)
sigaction(SIGTSTP, {sa_handler=SIG_IGN, sa_mask=SIG_DFL, sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x731f4e4a5c0b}, {sa_handler=SIG_DFL, sa_mask=[], sa
flags=0}, 3) = 0
open(AT_FDCWD, "/proc/self/maps", 0_RDONLY|O_CLOEXEC) = 3
printout(0, RLIMIT_STACK, NULL, (rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY)) = 0
fcntl(3, {it_mode=3_FMODE, it_size=0}, 0) = 0
read(3, "node[types]vnode/vtaps/vnode/vtd...", 1024) = 1024
read(3, "600af47d4000-000af48c1000 r-p 000000... 1024) = 1024
read(3, "00 00:30 13560 ... 1024) = 1024
read(3, "... /usr/lib/x86_64-linux... 1024) = 1024

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close(3)
stat(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=806, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, ({sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=&0x73f4ead5cb}, {sa_handler=SIG_GIG, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_SIGINFO, sa_restorer=&0x73f4ead5cb}), 0) = 0
rt_sigaction(SIGBUS, ({sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=&0x73f4ead5cb}, {sa_handler=&0x60afad0bc, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=&0x73f4ead5cb}), 0) = 0
rt_sigaction(SIGUSR, ({sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=&0x73f4ead5cb}, {sa_handler=&0x60afad0bc, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=&0x73f4ead5cb}), 0) = 0
ioctl(1, TCGETS2, &0xf9c9d3df80) = -1 EIOITY (Inappropriate ioctl for device)
busControlReset()
fcntl(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=34, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
stat(1, {st_mode=S_IFREG|0644, st_size=34, ...}) = 0
stat(1, {st_mode=S_IFREG|0644, st_size=1048576, ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 34
splice(4, NULL, 1, NULL, 34, 0) => call to Operating Systems Lib.
    = 34
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
fcntl(sock(fd), F_GETPIPE_SZ, 0, sa_flags=SS_DISABLE, ss_size=8192), NULL) = 0
munmap(&0x73f4ead5cb@0, 12288) = 0
exit_group(0) = ?
we exited with 0 sss
[shishir@kibinix: ~]$ ./practical25
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